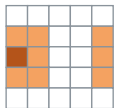


Spatial locality

encode the neighbourhood

transformer, Game of Life



grid (torus)



flattened:
neighbours scatter,
split at row ends

1D rotary positions



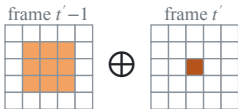
2D rotary positions



Temporal locality

store each observed transition

transformer, unseen rules



history token: cell at $t' \oplus$ its 3×3 at $t'-1$,
stored and recalled together

current cell only



+ previous-frame 3×3



Temporal stability

commit frames in order

diffusion model, Game of Life

joint



causal



joint denoising



causal freezing

